December 27, 2025

Building Government Services

Repeatable Services Algorithm

Introduction and Context

Government Services

Fragmentation and Redesign Challenges

Kenya's government service landscape is marked by **thousands of services** across national and county levels, leading to fragmented systems that hinder efficiency and effectiveness in service delivery.

The Core Problem

Government services are often treated as unique, leading to inefficiencies and disconnected systems, hindering effective service delivery.

Scale of Government Services

Service Diversity

Thousands of Services in Kenya

Kenya's government offers an estimated **4,000 to 6,000 services** across various sectors, highlighting the complexity and scale of service delivery within national and county frameworks.

Common Service Patterns

Government services typically adhere to **standardized flow patterns**, enhancing efficiency and consistency across various service types.

Pattern Type	Flow Example	Typical Use Case
Request → Review → Decision → Fulfilment	Citizen applies for a service	Licenses, permits, grants
Complaint → Assessment → Referral → Resolution	Report a problem	Helplines, ombudsman cases
Registration → Validation → Approval → Issuance	Register a new entity	ID cards, business registration
Application → Screening → Verification → Award / Rejection	Apply for scholarship or bursary	National bursaries, tenders
Intake → Classification → Routing → Handling → Closure	Case management	GBV/VAC cases, customer service desks
Notification → Investigation → Determination → Enforcement	Regulatory follow-up	Compliance, inspections
Data Capture → Approval → Disbursement	Payments, stipends	Pensions, subsidies
Incident → Response → Recovery → Reporting	Emergency handling	Disaster response, security incidents

Design Principles Overview

Government services should reuse the same core logic, changing only through configuration rather than custom code. This ensures scalability, maintainability, and faster service rollout.

Concept	Implementation
Algorithm	Constant across all services
Workflow	Defined per service through configuration
Rules	Adjust service behavior without changing code
Outcome	New services are created by **configuring rules & workflows**, not writing new software

Core Components:

- One Core Algorithm – the central engine driving all service logic
- One Workflow Engine – orchestrates steps, approvals, and notifications
- One Rules Engine – defines service-specific rules, validations, and decision criteria

Configuration Driven

Unifying Government Services Through One Approach

Emphasizing a **configuration over custom development** approach allows for a single algorithm, workflow engine, and rules engine to streamline service delivery across government entities.

The algorithm remains constant; services change through rules and workflow definitions

The Repeatable Government Service Processing Algorithm (GSPA)

All government services can be processed using a single, repeatable algorithm, ensuring consistency, efficiency, and scalability across ministries and counties.

Universal Steps:

- Service Intake – Capture requests, applications, or complaints
- Validation (Rules-based) – Apply service-specific rules to check eligibility, completeness, or compliance
- Automated Pre-processing (Optional) – Perform AI/automation tasks like verification, enrichment, or prioritization
- Workflow Routing – Direct the service to the correct department, officer, or queue
- Decision & Approval – Apply rules engine to approve, reject, or escalate the request
- Fulfilment – Deliver service outcome (document, payment, notification, or action)
- Closure & Feedback – Confirm completion and collect user feedback for improvement

High-Level Algorithm Flow

All services follow a standardized, configurable flow, allowing dynamic role assignment, rules-based validation, and automated output generation.

Step-by-Step Flow:

Citizen Selects Service – The user initiates a request or application

System Loads Service Configuration – Workflow and rules specific to the service are applied

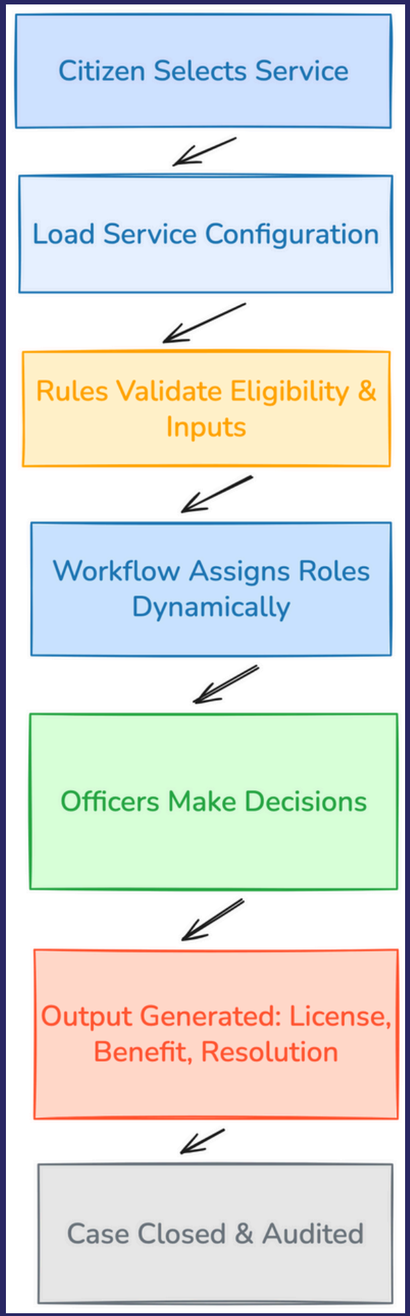
Rules Validate Eligibility and Inputs – Ensures completeness, accuracy, and compliance

Workflow Assigns Roles Dynamically – Tasks are routed to the appropriate officers or departments

Officers Make Decisions – Approve, reject, or escalate requests

Output is Generated – Service outcome is delivered (license, benefit, certificate, or resolution)

Case is Closed and Audited – Finalization, logging, and reporting for accountability



Operational Flow

Connecting Citizens to Government Services

The algorithm facilitates a seamless journey from citizen request to service resolution, ensuring efficiency and consistency across all government services through standardized processes and configurations.

Service Configuration Components

The interface allows for easy customization of eligibility rules, required documents, workflows, and approval processes to enhance service delivery.

Each government service can be fully defined through configuration, eliminating the need for custom code. This makes services repeatable, scalable, and easier to maintain.

- Eligibility Rules – Conditions defining who can access the service
- Required Documents – Documents or data inputs needed for processing
- Workflow Steps & Roles – Step-by-step process and assigned responsibilities
- Approval Hierarchy – Levels of authorization for decision-making
- SLA Timelines – Expected turnaround times for each step
- Output Type – Result delivered (certificate, license, payment, resolution, etc.)

Diverse Services, One Engine

A single core algorithm and workflow engine can process all government services, regardless of sector. The variability of services comes entirely from configuration, not custom code. This approach enables:

- Rapid rollout of new services without writing new software
- Consistency in citizen experience and service delivery
- Scalability across national and county levels
- Easier maintenance and updates, since the core engine remains unchanged

Examples of Government Services

Diverse Services Using One Engine

- Birth Registration – Civil registry services
- Trade Licensing – Business permits and licensing
- GBV / VAC Case Management – Child and gender protection services
- Data Protection Complaints – Regulatory complaint handling
- Social Protection Enrollment – Benefits, bursaries, and subsidies
- County Inspections – Health, safety, and regulatory inspections

Benefits of a Repeatable Algorithm

Standardization

Achieving consistency across government services

Consistent Interfaces

Standardized interfaces across services ensure that users have a **familiar experience**, enhancing accessibility and reducing the learning curve for citizens engaging with multiple government services.

Streamlined Processes

Uniform protocols and workflows promote **efficiency**, enabling quicker service deliveries and minimizing errors, which leads to improved citizen satisfaction and trust in government systems.

Resource Optimization

Reducing fragmentation allows government resources to be **allocated more effectively**, focusing efforts on improvement and innovation rather than duplicating efforts across different service delivery platforms.

Enhancements

Key improvements in government service delivery

Enhanced Auditability

The algorithm facilitates **robust audit trails**, ensuring transparency at every stage of service delivery, allowing for thorough evaluations and accountability across various government operations.

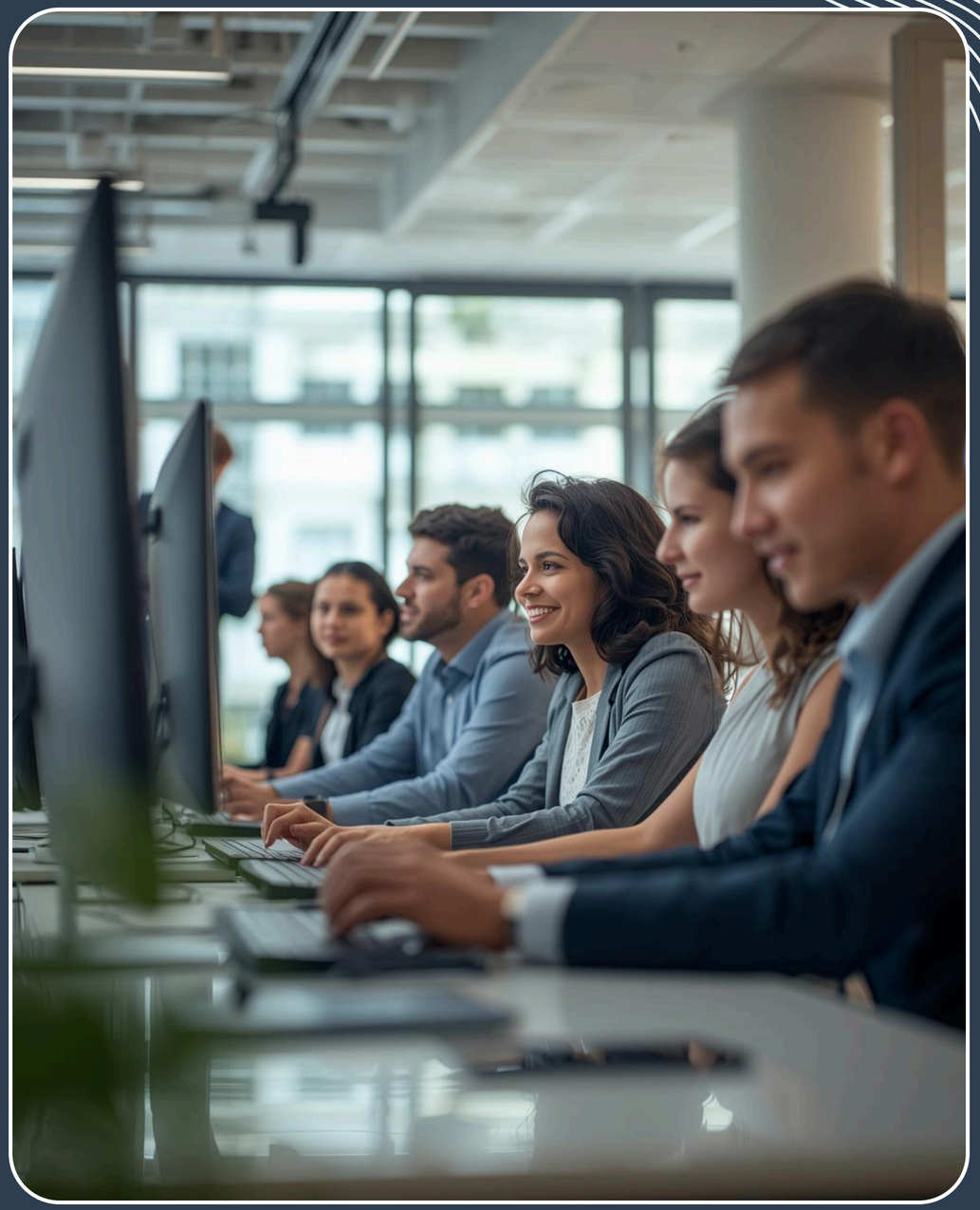
Simplified Policy Updates

Centralized configurations allow for quick modifications of policies and processes, streamlining the adaptation to new regulations without the need for extensive system overhauls or disruptions.

AI Readiness

The standardized approach positions government services to leverage **artificial intelligence**, enabling predictive analytics and automation to enhance service efficiency and responsiveness to citizen needs.

Alignment with Digital Agenda



eCitizen Support

Enhancing Digital Government Service Access

The eCitizen platform streamlines access to essential government services, promoting efficiency and transparency while fostering **citizen engagement** across national and county levels in Kenya.

Strategic Impacts

Embracing digital services reduces ICT costs while enhancing overall citizen experience and satisfaction with government interactions.

Workflow Engines

Integrating **workflow engines** with BPMN standards enhances interoperability, enabling efficient service delivery across government platforms.

Conclusion and Call to Action

Thank You

Phone
123-456-7890

Email
hello@reallygreatsite.com

Social
@reallygreatsite